

ECG Image Digitization

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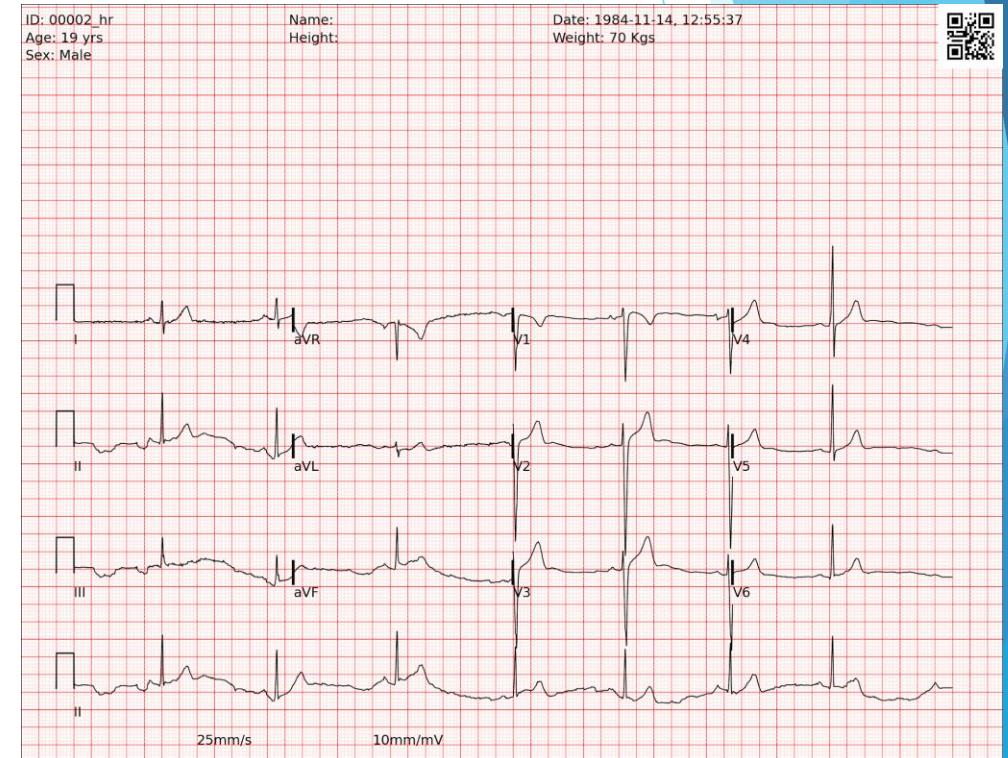
Background

- ▶ Electrocardiogram (ECG)
 - Records electrical activity of the heart
 - Helps with diagnoses of heart rhythm diseases and heart attacks
- ▶ ECGs are often printed on ECG thermal paper
 - Reliable, Standardized, Widely Used
 - Still used for a variety of reasons:
 - Familiarity
 - Convenience
 - Ease of Interpretation
- ▶ Increased need for digitalization
 - Digital/Electronic health records & storage
 - Potential support for AI/ML analysis



Generate Distortionless Image

- ▶ Pick Size and Resolution
- ▶ Plot ECG
- ▶ Add grid lines
 - ▶ Made up of 1mm and/or 5mm side length squares
 - ▶ Each 1mm vertically is 0.1mV, horizontally is 50ms
- ▶ Add header and labels
 - ▶ Header: Date, Sex, Age, etc.
 - ▶ Labels: Number ECGs, mark points in time

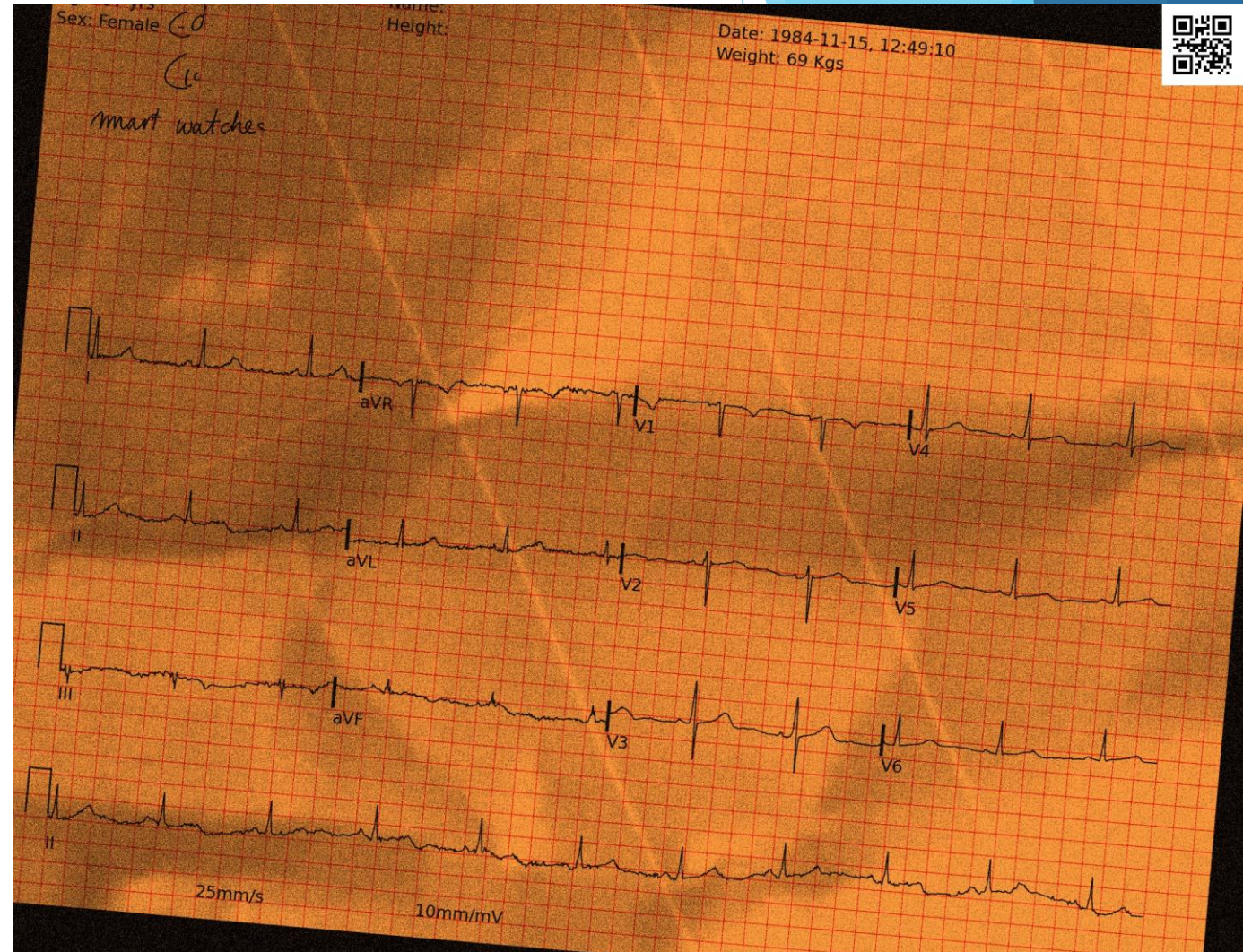


Apply Distortions

The distortion effects in ECG-Image-Kit are:

- ▶ Handwritten Notes
- ▶ Wrinkles
- ▶ Creases
- ▶ Rotation
- ▶ Crop
- ▶ Noise
- ▶ Color Temperature

All are applied in this image, ask if any are unclear



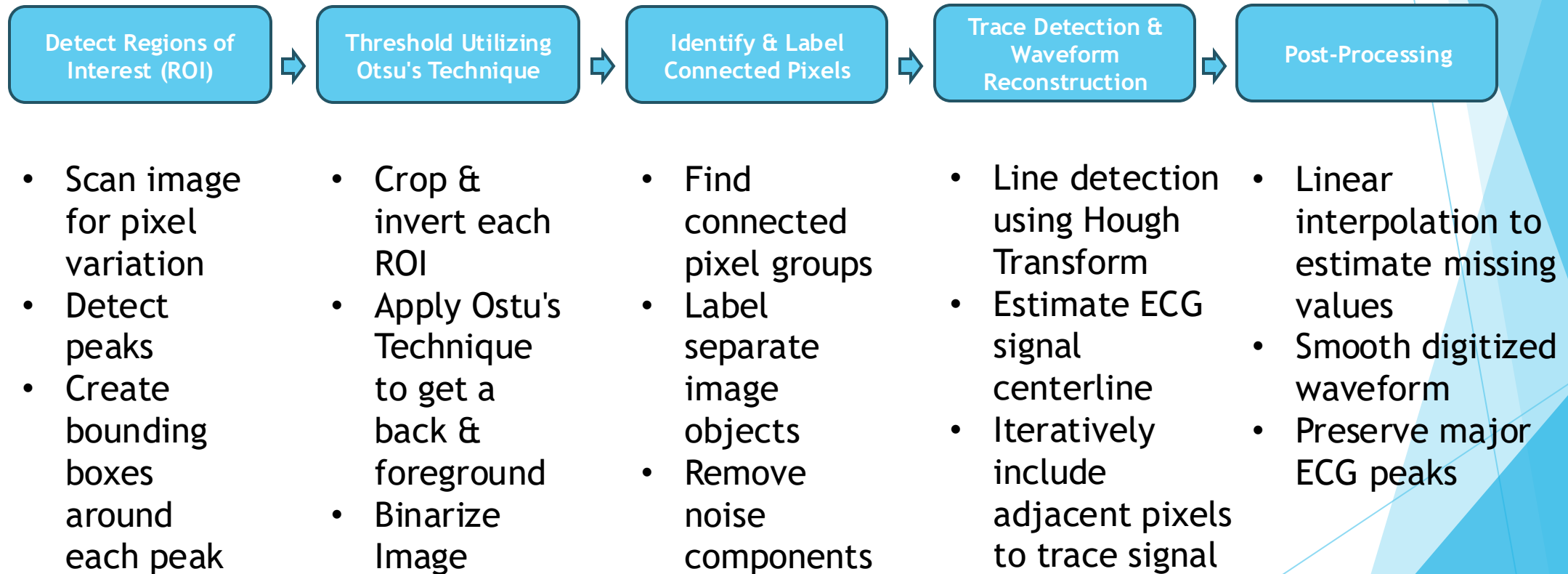
Limitations of ECG-Image-Kit

- ▶ No method for image compression artifacts
- ▶ Generate unique wrinkle patterns
 - ▶ Wrinkles are generated using image quilting on a set of 20 maps
- ▶ Water damage/blurring ink
- ▶ Perspective
 - ▶ “Camera” can only be parallel to image

All of these would not be strange to see in a real image of a paper ECG and can't be replicated with this toolkit used by many existing methods.

Baseline Algorithm Replication

Replicating image processing and digitization pipeline from Baydoun et al. (2019)



Expected Results & Benchmarks

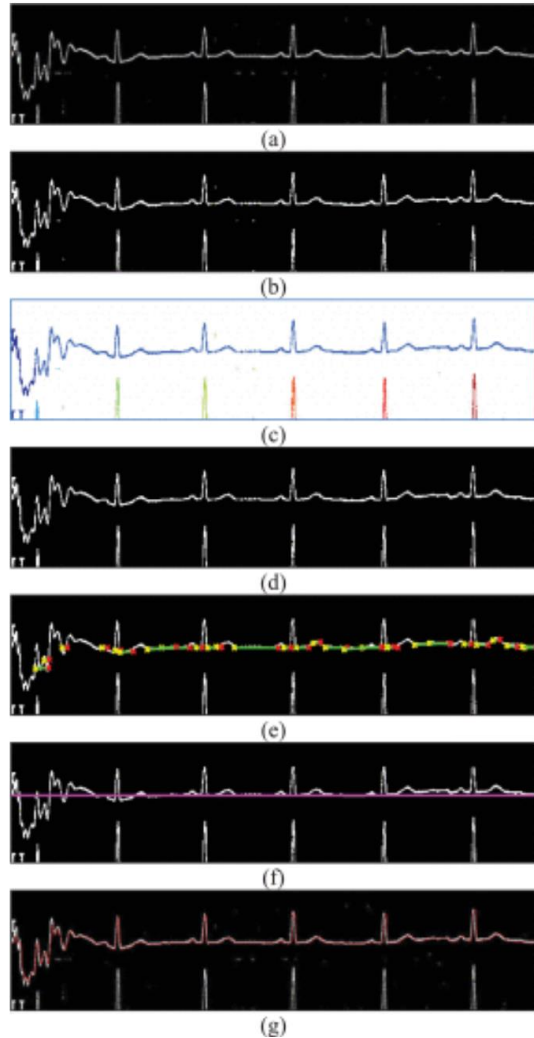
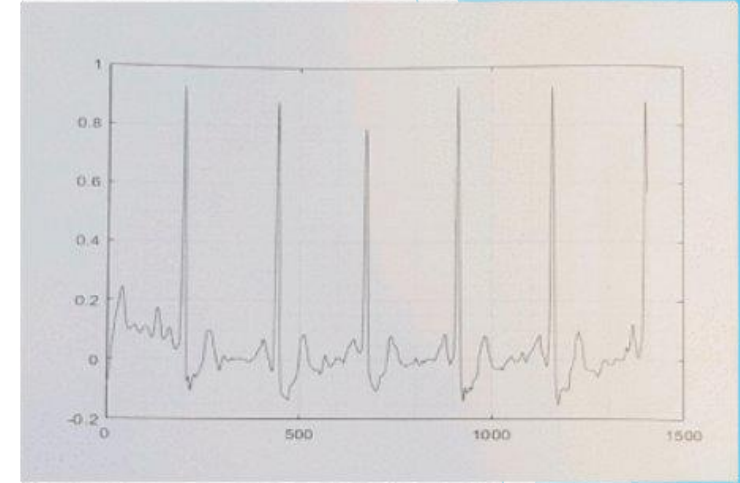
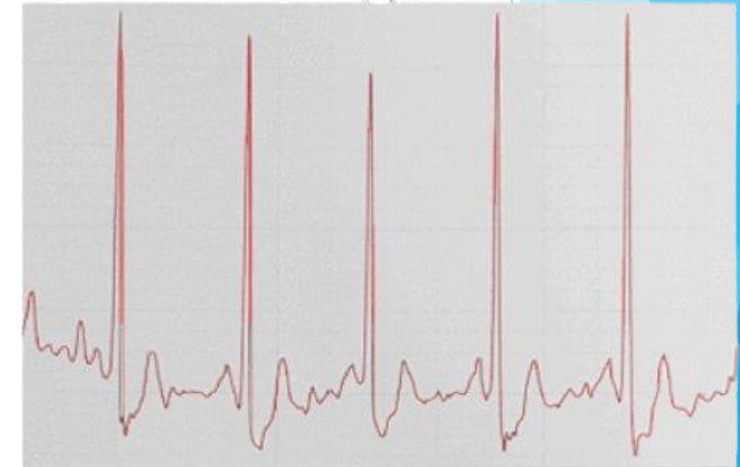


Figure 2: Expected Extraction Pipeline

- Successful extraction of ECG signal from scanned images
- Preservation of major peaks and overall waveform
- Compare reconstructed signal with reference ECG
 - Determination of Pearson's Linear Vector Correlation Coefficient
 - Avg. Published value of 0.952 ± 0.036 across 30 ECGs



(a)



(b)

10/6/2026

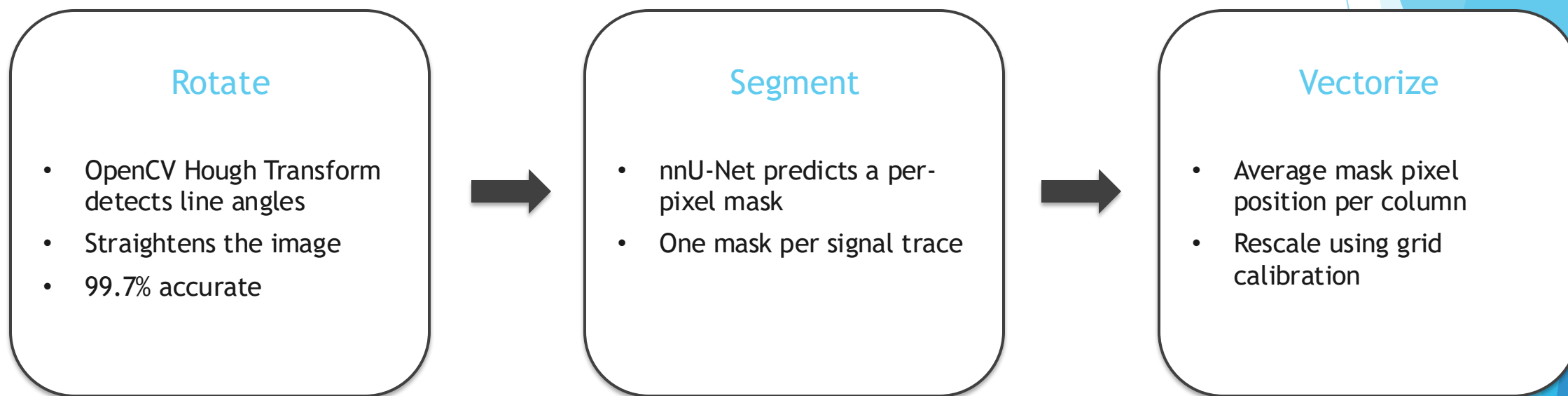
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Figure 3: Expected Digitized Output

SignalSavants - University of Oxford

PhysioNet Challenge 2024 • Digitization task: SNR 12.15, 1st place of 16 teams

Three-stage pipeline of rotations, segmentation, and vectorization



- Training images were self-generated from PTB-XL via ECG-Image-Kit, with 4+ augmented variants per signal
- Their winning model used only 20k images and simpler masks. Larger, more complex models overfit

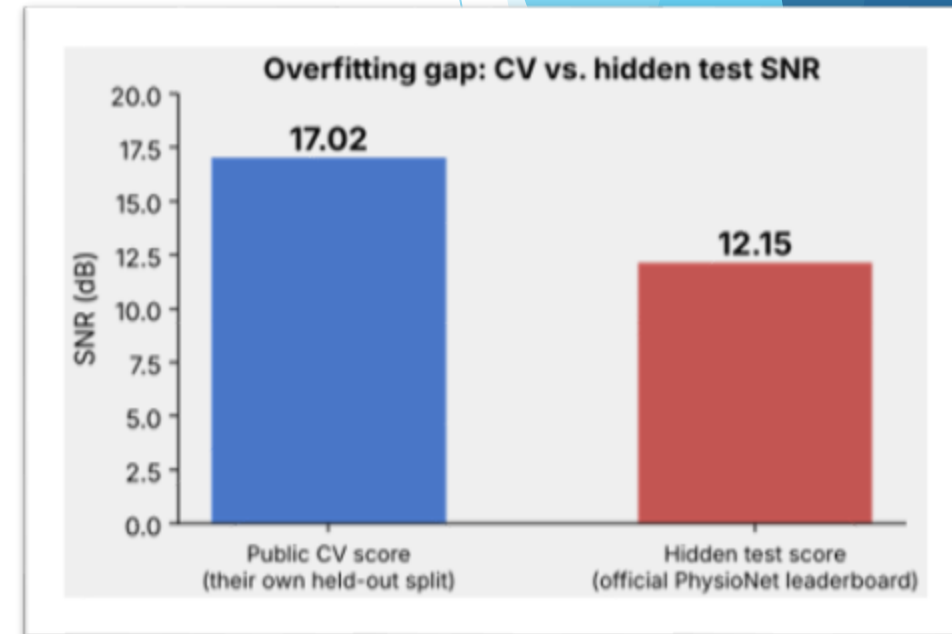
SignalSavants - University of Oxford

► Strengths:

- Classical computer vision handles rotation while deep learning handles the harder segmentation step.
- Best performance overall on the PhysioNet challenge, can be considered “state of the art” in a sense.
- Code and weights are open-source to provide natural baseline.

► Weaknesses:

- Fails on image types outside training, giving negative SNR on phone photos and screenshots, despite working on paper scans.
- Big gap between CV score (17.02) and hidden-test score (12.15) suggests overfitting to their synthetic image style.



WAVIE Method - University of Auckland

PhysioNet Challenge 2024 • Digitization task: SNR 5.469, 3rd place of 16 teams

Separate modules for rotation correction, signal segmentation, lead detection, and signal digitization.

Rotation Correction

Deep Image Orientation
Angle Detection

- Predict rotation angle θ
- Train with $\theta \sim N(0, 25.2^2)$, emphasizing realistic small rotations
- Apply inverse angle $-\theta$
- Output: standardized image orientation

Signal Segmentation

2D nnU-Net

- Input: corrected ECG image
- Output: binary mask of all lead signals
- 250 epochs; initial LR = 0.01 with polyLR
- Augmentations: rotation, downsampling, mirroring

Lead Localization

YOLOv5s

- Detect signal bounding boxes and lead-name boxes
- Pretrained weights; trained 300 epochs
- Post-process to retain 13 boxes: 12 leads + 1 rhythm lead

Signal Digitization

OpenCV + signal processing

- Crop mask using YOLO boxes
- Extract contours
- Column-wise median; one y-position per x
- Pixel $\rightarrow$ time/voltage (0.04 s / 0.1 mV)
- Sliding-window baseline estimate

- Training images were self-generated from PTB-XL via ECG-Image-Kit, different configurations were made
- Five generation configurations were used for training and ten for the held-out data
- 6 of 10 held-out generation configurations were unseen during training

WAVIE Method - University of Auckland

▶ Strengths:

- ▶ Modular 4-stage pipeline is easy to debug module-by-module
- ▶ Explicitly localizes all 12 leads + rhythm strip, not single trace
- ▶ Train/held-out split tests generalization
- ▶ Rotation model trained on realistic small-angle distortions ($\theta \sim N(0, 25.2^\circ)$)

▶ Limitations:

- ▶ Far lower SNR than SignalSavants (5.469 vs. 12.15)
- ▶ Heavy dependency on YOLO lead detection, as a bad box corrupts everything downstream
- ▶ Digitization assumes one trace per column, breaks on overlapping/crossing leads

Next Steps

- ▶ Test ECG-Image-Kit tools to experiment with ECG image generation
- ▶ Create dataset for initial experimentation
- ▶ Reconstruct MATLAB baseline algorithm
- ▶ Test code from SignalSavants and WAVIE

Thank you for listening

Any Questions?

References

- [1]: Baydoun, M., Safatly, L., Abou Hassan, O. K., Ghaziri, H., El Hajj, A., & Isma'eel, H. (2019). High Precision Digitization of Paper-Based ECG Records: A Step Toward Machine Learning. IEEE journal of translational engineering in health and medicine, 7, 1900808. <https://doi.org/10.1109/JTEHM.2019.2949784>
- [2]: Krones, J. S., Walker, W., Lyons, T., & Mahdi, A. (2024). Combining Hough Transform and Deep Learning Approaches to Reconstruct ECG Signals From Printouts. arXiv:2410.14185.
- [3]: Verlyck, M. A., Dillon, J. R., Creamer, S. A., & Zhao, D. (2024). WAVIE: A Modular and Open-Source Python Implementation for Fully Automated Digitisation of Paper Electrocardiograms. Computing in Cardiology, 51. DOI: 10.22489/CinC.2024.229.